

Transmitter

Information bits

1 0
1 1

(8,8)-16APSK modulation



amplitude factor

Block fading



Atmospheric FSO channel

Carrier phase

CFO · Doppler · PN

Complex AWGN



phase

additive

Composite FSO channel



Channel blocks
100 symbols

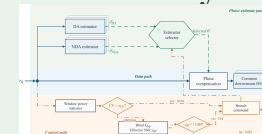
DSP window
256 samples

Within one DSP window



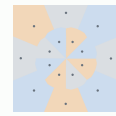
Coherent receiver DSP

Adaptive carrier recovery



Detailed in Fig. 2

16APSK demodulation



Detected bits